Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/26/2024
PI Name:	Sargsyan, Ashot M.D.		
Project Title:	Imaging-Based Human Physiology Research and Innovation in Early Spaceflight: Inspiration4 Mission		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	asargsyan@wylehou.com	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	2400 NASA Pkwy Mail Code PMO		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	05/01/2021	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	SpaceX		
Flight Assignment:	SpaceX Inspiration4 NOTE: End date changed to 09/30/2025 per TRISH (Ed., 8/7/24). NOTE: End date changed to 04/30/2024 per TRISH (Ed., 8/4/22). NOTE: End date changed to 04/30/2023 per TRISH (Ed., 8/4/22). Project formerly titled, "Miniaturized Medical Imaging for Space: Butterfly iQ+ Ultrasound".		
Key Personnel Changes/Previous PI:	Douglas Ebert, Ph.D. has been added as a Co-Investigator (Ed., 1/6/24).		
COI Name (Institution):	Ebert, Doug Ph.D. (KBR)		
Grant/Contract No.:	NNX16AO69A-EXP0002		
Performance Goal No.:			

Performance Goal Text:	NOTE: End date changed to 09/30/2025 per TRISH (Ed., 8/7/24).
Task Description:	EXPAND Project This proposal aims to perform a series of low-resource, high-yield human research objectives using a hand-held ultrasound imaging system Butterfly iQ+ (Butterfly Network, USA) during the commercial mission Inspiration4: 1) urinary bladder function in microgravity; 2) major vein and fluid status evaluation; 3) ocular imaging self-surveillance protocol; and 4) gallbladder function in microgravity. These objectives address recognized concerns for space flight health, with parallel consideration of benefits to certain patient populations, including pediatric malignancies. They are transferable, scalable, and will establish precedents for long-term inquiries. Traditional imaging-based research relies heavily on crew training, infrastructure, and ground personnel; it foregoes science opportunities that require quick deployment, simple procedures, flexible schedules, and crew autonomy. This proposal leverages the inherent features of the Butterfly iQ+ ultra-compact ultrasound imager, the team's experience with hybrid crew skill management solutions, and the nature of the Inspiration4 mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results generated during these experiments will benefit the realms of 1) spaceflight physiology and 2) skill management. Physiological elements of the project will add to space medicine and physiology knowledge base by providing data about 1) urinary bladder function, 2) internal jugular vein flow patterns, and 3) ocular structure changes. The data collected during this project will be unique given that they are collected in the early phases of spaceflight exposure (first 3 days), as compared with the majority of available data which have been collected much later in flight (30+ days for ISS missions). In the skill management realm, this project will have application for remote medicine, spaceflight medicine, home care (telemedicine), and general medical training literature. Skill management solutions will be developed and testing that will directly benefit programs requiring that minimally-trained operators perform complex tasks and are particularly applicable to situations where communication is limited either in quality or quantity. Latency and bandwidth considerations are central to exploration spaceflight and a variety of Earth based scenarios where increased autonomy is necessary. This work contributes to the academic knowledge base in the training literature but will simultaneously be building operational experience and products for both NASA and commercial space operators.
Task Progress:	This summary consists of three sections, each representing a finite project funded under the original proposal: A. Original Objectives of Inspiration4 Mission; B. Polaris Dawn; C. AAD A. Original Objectives of Inspiration4 Mission: This project performed low-resource, high-yield human research objectives addressing known concerns for space flight health, using a Butterfly iQ+ handheld ultrasound system (Butterfly Network, USA) during the commercial mission Inspiration4: 1) urinary bladder function in microgravity; 2) major vein and fluid status evaluation; and 3) ocular imaging self-surveillance protocol. This research leveraged features of an ultra-compact ultrasound imager, the team's experience with hybrid skill management solutions, and the nature of the Inspiration4 mission. Given very limited preflight training (by proxy) and full crew autonomy, we relied on just-in-time (JIT) instructions to impart conceptual and procedural information. Given the risks to data quality due to lack of imaging expertise, and to evaluate the JIT tools with objectivity, a formalized data quality assessment was conducted before data analysis. This innovation included scoring of each image for anatomical accuracy and technical quality on a scale of 0 – 3 (0 = no useful data, 3 = highest achievable score) and setting a data usability threshold at 2.0. Urinary bladder imaging scores showed effectiveness of the system for autonomous operation. However, physiological trends remained unclear due to varied timing of data relative to voiding. Internal jugular vein (IJV) imaging was used to identify or rule out anomalies (thrombi, stasis, flow reversal, spontaneous echo contrast (SEC)). IJV images had highest quality scores. Spontaneous antegrade flow was demonstrated and no SEC was detected. Impedance Threshold Device (ITD) reduced right IJV area by an average of 36% preflight (supine) and 29% inflight. In-flight images from ocular procedure (most intricate of three) scored below the usability threshold in all subjects. Therefore, ocular imaging data were deemed inadequate for assessing microgravity-induced structural changes. Thus, minimally trained crew, with the aid of JIT materials, succeeded with 2 out of 3 procedures. Quality of images, expressed as success scores, depends on the complexity of the autonomous procedures. The guidance for ocular procedure, which requires strict positioning and gaze control, was deemed insufficient when coupled with virtual lack of preflight training. This highlights the need to ensure mental model development for the key demands of each procedure. The two successful procedures demonstrated a significant medical and research capability in situations with degraded communications. As a valuable insight, IJV evaluations did not reveal flow anomalies in any subjects within the early exposure timeframe. If corroborated in the Ultrasound Innovation study, this finding may help elucidate progression patterns and mechanisms of flow degradation in the left IJV. As an important "first", ITDs were flown in this mission and used by all crewmembers for hemodynamic modification during IJV testing. The results of the experiment are published in: Jones CW, Overbey EG, Lacombe J, et al. Molecular and physiologic changes in the SpaceX Inspiration4 civilian crew. Nature. 2024 Jun 11. doi: 10.1038/s41586-024-07648-x . PMID: 38862026. B. Ultrasound Innovation (Polaris Dawn mission): Completed all preflight activities; launch NET July 12, 2024. This proposal will use Butterfly iQ+ ultrasound during Polaris Dawn to pursue the following objectives: 1) urinary bladder function in microgravity; 2) major vein evaluation with use of ITD; 3) bubble detection/Venous Gas Embolism (VGE) evaluation; 4) Effects of lower extremity blood flow restriction on central (IJV) and peripheral (popliteal vein) circulation. Training and execution approaches are similar to Inspiration4. In April-May, 2024, another objective was added in synergistic collaboration with the Blood Flow Restriction (BFR) experiment (Objective 4). Two of the four objectives are contiguous with Inspiration4 to generate larger data sets from early microgravity exposure. Each

	objective, in addition to physiological data, will pursue refinement and objective evaluation of autonomy solutions. The IJV evaluation component will rely on a procedure guide produced by Level Ex, thus leveraging existing expertise to innovate in procedural performance in autonomy. Due to launch delays, a second crew training and BDC session were conducted in proximity of launch. The mission is planned to launch NET July 12, 2024. C. Butterfly iQ+ Antarctic Program: Exploring Autonomous Procedures - data collection (imaging data sets and questionnaires) in progress. This is a spaceflight analog study performed in 4 small Australian Antarctic research stations. The aims are to :1) Determine the effectiveness of just-in-time (JIT) training materials for specific ultrasound scanning protocols with terrestrial and space health relevance, using non-sonographer operators of different backgrounds; 2) Compare autonomously acquired image quality metrics between medical and non-medical personnel: 3) Evaluate autonomously acquired image quality metrics during the course of a 5-month mission. Each test session includes imaging procedures of different complexity guided by operator-paced linear JIT instructions. Analysts will be completely blinded. As of this APR, >178 imaging protocols have been completed. This analog study will combine with the previous 2 flight studies to produce a multi-faceted report on the multiple aspects of autonomous medical procedures.
Bibliography Type:	Description: (Last Updated: 05/19/2026)
Articles in Peer-reviewed Journals	Jones CW, Overbey EG, Lacombe J, Ecker AJ, Meydan C, Ryon K, Tierney B, Damle N, MacKay M, Afshin EE, Foox J, Park J, Nelson T, Suhail M, Byhaqui SG, Aslam B, Tali UA, Nisa L, Menon P, Patel CO, Khan SA, Ebert DJ, Everson A, Schubert MC, Ali NN, Sarma MS, Kim J, Houerbi N, Grigorev K, Garcia Medina S, Summers AJ, Gu J, Altin JA, Fattahi A, Hirzallah MI, Wu JH, Stahn AC, Beheshti A, Klotz R, Ortiz V, Yu M, Patras L, Matei I, Lyden D, Melnick A, Banerjee N, Mullane S, Kleinman A, Loesche M, Menon AS, Donoviel DB, Urquieta E, Mateus J, Sargsyan AE, Shelhamer M, Zenhausern F, Bershad EM, Basner M, Mason CE. "Molecular and physiological changes in the SpaceX Inspiration4 civilian crew." Nature. 2024 Jun 11. https://doi.org/10.1038/s41586-024-07648-x ; PMID: 38862026 , Jun-2024